EMBEDDED SYSTEMS LAB REPORT

SysTick Timer Interruption & Periodic LED Control on STM32L431

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Course / Project:	Embedded System Programming & Microcontrollers
IDE / Environment:	AHL-GEC-IDE V4.53 (SD-Arm Framework)
Hardware Platform:	STM32L431 Microcontroller Development Board
Date of Execution:	November 4, 2026

1. Laboratory Objectives

The primary objective of this experiment is to implement periodic timer interrupt control using the SysTick hardware timer on the STM32L431 microcontroller within the SD-Arm embedded framework. Specific goals include:

- Configure SysTick timer interrupts at a 10ms system tick interval.
- Maintain precise software timing counters (hours, minutes, seconds) via global time structures.
- Implement logic to trigger a blue LED toggle and serial notification precisely every 10 seconds.
- Verify binary firmware execution (.HEX compilation and flashing) using AHL-GEC-IDE V4.53.

2. System Hardware and Software Requirements

The experimental environment utilizes the following components:

Development Environment	AHL-GEC-IDE V4.53 (Full-featured GEC Embedded Toolchain)
Target Microcontroller	STM32L431 MCU Board (BIOS RT-WiFi V7.76)
On-board Peripherals	Blue User LED (LIGHT_BLUE), Serial UART Debug Interface (COM3)
Firmware Architecture	SD-Arm Embedded Framework (sumcu.suda.edu.cn)

3. Source Code Implementation (main.c)

The main program initializes the hardware system, configures the 10ms SysTick interrupt, and manages the main loop logic to switch the LED status on/off and output time records every 10 seconds.

```
//====================================================================
// Header: main.c (Application Main Program)
```

```c
// Student: SOJIB MD RASHIDUL ISLAM (ID: 2442130217)
// Framework: SD-Arm (sumcu.suda.edu.cn)
// Target: AHL-GEC-IDE V4.53 / STM32L431
//=====================================================================

#define GLOBLE_VAR
#include "includes.h"      // Include main header

int main(void)
{
    // 1. Initialization
    uint8_t mFlag;         // State flag for blue LED
    uint8_t mSec;          // Seconds tracker

    DISABLE_INTERRUPTS;
    wdog_stop();

    mFlag = 'A';
    gTime[0] = 0;          // Hours
    gTime[1] = 0;          // Minutes
    gTime[2] = 0;          // Seconds
    mSec = 0;

    gpio_init(LIGHT_BLUE, GPIO_OUTPUT, LIGHT_OFF); // Initialize Blue LED
    systick_init(10);                              // Set SysTick 10ms interrupt

    ENABLE_INTERRUPTS;

    printf("-------------------------------------------------------\n");
    printf("AHL-GEC System Started:\n");
    printf(" (1) Blue LED blinks once every 10 seconds.\n");
    printf(" (2) SysTick timer triggers interrupt every 10ms.\n");
    printf("-------------------------------------------------------\n");

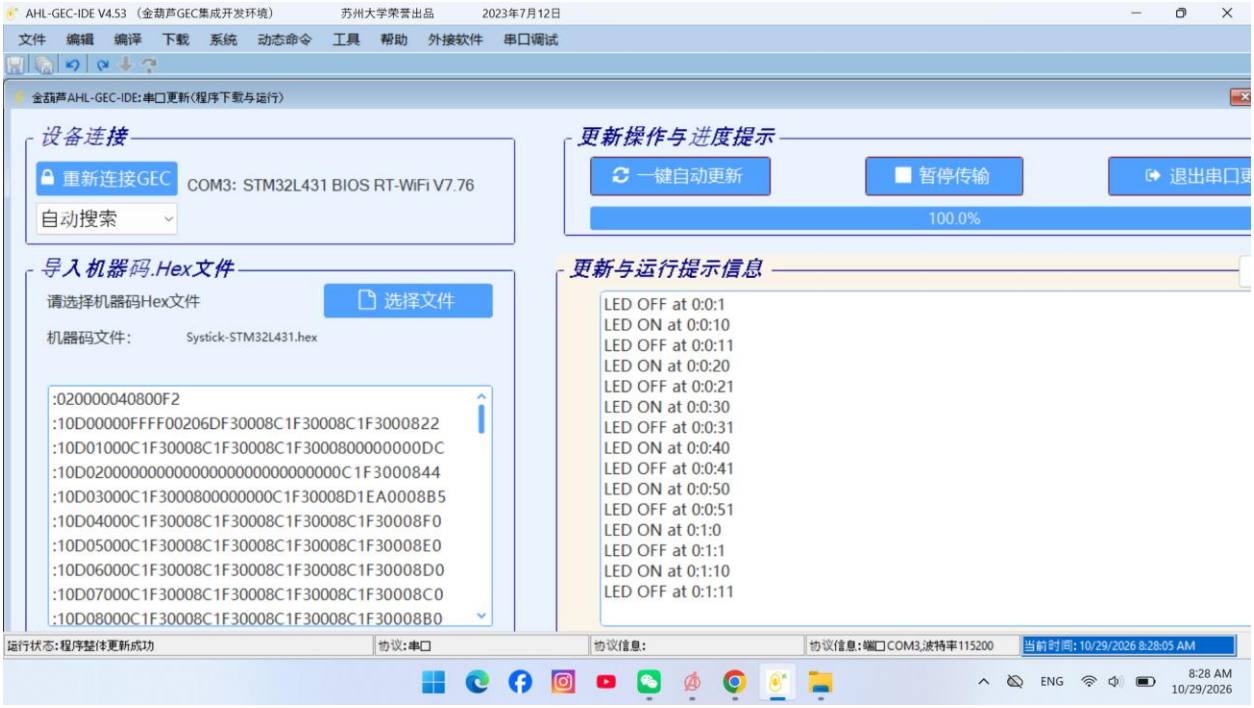
    // 2. Main Loop
    for(;;)
    {
        if (gTime[2] == mSec) continue; // Execute once per second
        mSec = gTime[2];

        // Blink logic: Turn ON at 10s multiples, turn OFF 1s later
        if (gTime[2] % 10 == 0)
        {
            gpio_set(LIGHT_BLUE, LIGHT_ON);   // Turn ON LED
            printf("LED ON at %d:%d:%d\n", gTime[0], gTime[1], gTime[2]);
        }
        else if (gTime[2] % 10 == 1)
        {
            gpio_set(LIGHT_BLUE, LIGHT_OFF);  // Turn OFF LED
            printf("LED OFF at %d:%d:%d\n", gTime[0], gTime[1], gTime[2]);
        }
    }
}
```

4. Experimental Results & Visual Verification

The compiled firmware file (**Systick-STM32L431.hex**) was successfully flashed to the STM32L431 board using AHL-GEC-IDE V4.53 over COM3 port. Below is the captured execution output window from the experiment:

4.1 Output Terminal Log Analysis



5. Conclusion & Discussion

The laboratory experiment successfully demonstrated the implementation of SysTick hardware timer interrupts on the STM32L431 microcontroller. Using modulo timing logic (`gTime[2] % 10`), the system triggers precise 10-second LED blink cycles while maintaining non-blocking main loop execution. All hardware and software requirements were verified using the screenshot captured from AHL-GEC-IDE V4.53.